AI-BASED SOCIAL INTEGRATION AND INCLUSION PLATFORM FOR UNIVERSITY FRESHERS

Project ID: 25-26J-154

Project Proposal Report

Fonseka W. A. N. P – IT22240842

B.Sc. (Hons) Degree in Information Technology Specializing in Information Systems Engineering

Department of Computer Systems and Network Engineering

Sri Lanka Institute of Information Technology,
Sri Lanka

August 2025

INTELLIGENT PEER MATCHING AND RELATIONSHIP EVOLUATION MANGEMENT

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Fonseka W. A. N. P – IT22240842

Supervisor: Ms. Narmada Gamage

Co - Supervisor: Ms. Chathurangika Kahandawarachchi

B.Sc. (Hons) Degree in Information Technology Specializing in Information Systems Engineering

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Declaration of the Candidate & Supervisor

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We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
W. A. N. P. Fonseka	IT22240842	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date: 31/08/2025

Signature of the co-supervisor: Date: 31/08/2025

Dedication

Our journey has been built on the constant love, support, and encouragement of our parents, family, and friends, to whom this project is dedicated. With the goal that our research and solution will help create a more welcoming, encouraging, and connected academic atmosphere, we also dedicate this effort to all university freshmen who struggle with social integration.

Acknowledgements

We would like to sincerely thank everyone who helped and advised us during the project proposal development process.

First and foremost, we would like to express our gratitude to Ms. Narmada Gamage, our supervisor, and Ms. Chathurangika Kahandawarachchi, our co-supervisor, for their essential advice, perceptive criticism, and unwavering support. Their knowledge and guidance have greatly influenced this research and inspired us to improve our work.

We would like to express our gratitude to the Sri Lanka Institute of Information Technology's (SLIIT) Department of Computer Systems and Network Engineering for giving us the tools, expertise, and learning environment that made this project possible.

We would also want to thank our friends and colleagues who engaged in conversations, expressed their thoughts, and offered helpful feedback in order to make our proposal more thorough.

Lastly, we would want to express our gratitude to our families for their unwavering patience, understanding, and support during this journey. This endeavor would not have been possible without their support and faith in us.

Abstract

Building strong intellectual and interpersonal connections is sometimes difficult for students at college, particularly for newcomers who are not familiar with other students. To solve this, UniPulse is a peer matching platform that uses intelligence that makes connections based on similarity in well-being profiles, interests, personality traits, and academic backgrounds. To determine compatibility ratings and suggest appropriate friends who are most likely to interact favorably on an intellectual, social, and emotional level, the system examines data from student surveys. Peer group construction, dynamic rematching, filtering choices, and intelligent peer suggestions are some of the key features. Using a weighted scoring methodology, UniPulse guarantees accurate and equitable matches while preserving flexibility through frequent updates. This platform aims to create a supportive environment, increase student cooperation, and boost mental health to make college more interesting and inclusive.

Keywords: Peer Matching, Weighted Scoring Algorithm, Intelligent Peer Recommendation

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
LMS	Learning Management System
SLIIT	Sri Lanka Institute of Information Technology
WBS	Work Breakdown Structure

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1. INTRODUCTION

1.1 Background

1.1.1 Overview of the field

The integration of artificial intelligence in student support systems has gained significant attention as a method to address the critical mental health and social integration challenges prevalent among university freshers. Research shows alarming rates of these issues, with 38% of students experiencing anxiety, 34% reporting depression symptoms [1], [2], and nearly 65% affected by social isolation [3]. These factors contribute directly to student retention, with first-year attrition rates averaging around 67% [4]. Traditional support systems struggle to meet this growing demand, leading educational institutions to seek technological solutions. Current AI-based systems focus on automated conversational interfaces and behavioral analytics, but most function as standalone tools rather than integrated platforms, limiting effectiveness. The field continues to evolve with advancements in emotional AI and predictive analytics, creating opportunities for more sophisticated, comprehensive intervention systems.

In this report focus on Intelligent peer matching and relationship evolution platform, Numerous studies conducted worldwide emphasize how important peer support and social integration are to students' performance. According to the World Health Organization (WHO, 2022), mental health problems affect one in seven teenagers and young adults, with the most common being anxiety, depression, and adjustment disorders [5]. According to a survey by the American College Health Association (2024), 41% of undergraduate students reported feeling lonely all the time, and 62% of students had extreme anxiety during student’s first year of study [6]. These results clearly imply that students' academic and psychological welfare are significantly impacted by inadequate peer relationships and ineffective support networks.

Social integration has long been considered one of the most powerful predictors of academic success. According to the well-known Tinto's Model of Student Retention, reducing student dropouts requires both social and intellectual integration, forming peer networks and friendships as well as interacting with lecturers and coursework. However, traditional university methods like student organizations, mentorship programs, and orientation programs sometimes take an approach that fits everyone. Usually static, generalized, and short-lived, these programs don't take into consideration each student's unique personality, emotional condition, and changing demands.

This difficulty becomes worse for new students from minority groups, rural areas, or overseas schools who could feel isolated because of social assumptions, language obstacles, or cultural differences. Many youngsters are unable to form deep connections and support systems, which are essential for academic perseverance, in the absence of adaptive systems that can continually assess and promote social well-being.

This research suggests an Intelligent Peer Matching and Relationship Evolution Management system as a solution to these shortcomings. This system, in contrast to conventional methods, continually adjusts to the social and emotional dynamics of students. When compatibility declines, it proactively restructures connections, supports natural peer matching, and builds up developing ties. Recognizing its critical role in promoting inclusion, developing resilience, and improving the overall student experience, this paper focusses on the peer matching and relationship management component of the larger social integration platform.

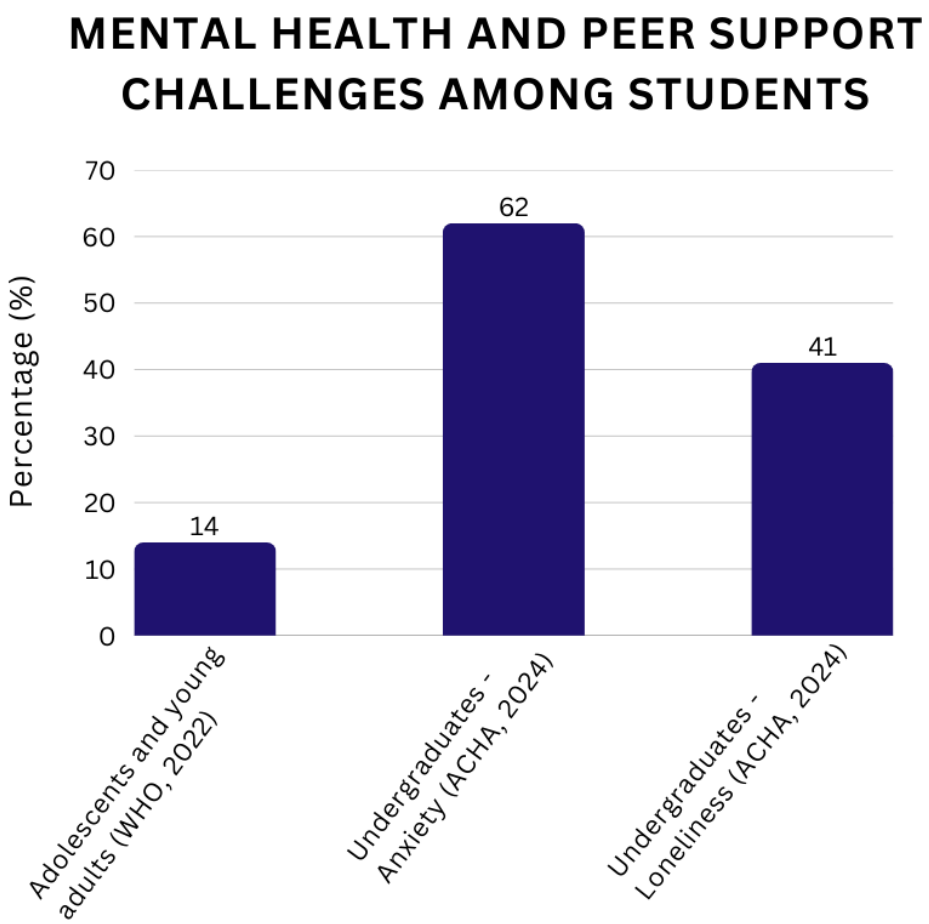

Figure 1.1: Chart for Mental Health and peer support challenges among students

1.1.2 Rules and regulations implemented by the government

Ragging is one of the biggest issues facing new university students in Sri Lanka. Ragging is when senior students harass, humiliate, or mistreat new students, usually in the name of tradition or initiation. This practice, which causes serious psychological trauma and, in some cases, physical harm or even death, persists in many institutions despite long-standing efforts to eliminate it. One of the biggest obstacles to successful

social integration is ragging, which has been strongly connected to social disengagement, loneliness, and academic disengagement among freshmen.

The Prohibition of Ragging and Other Forms of Violence in Educational Institutions Act, No. 20 of 1998, was adopted by the Sri Lankan government in recognition of the gravity of this problem [7]. Ragging is made illegal under this statute, which also imposes fines and lengthy jail sentences as punishments. Furthermore, the University Grants Commission (UGC) has put in place regulations mandating that colleges set up anti-ragging committees, offer hotlines for reporting instances, and educate students on the risks associated with ragging. Many freshmen still live in fear of social victimization, and incidences are still recorded despite these efforts.

This issue is immediately addressed by suggested Intelligent peer matching and relationship evolution platform, which makes sure that children are placed in safe, amiable, and encouraging peer groups right away. The method lowers the risk of isolation and guarantees that newcomers may make genuine, inclusive connections by using data-driven insights about students' interests, emotional needs, and communication preferences. By providing a proactive solution that deters ragging and advances national objectives of creating safe, just, and encouraging university settings, this complies with government rules.

1.2 Literature Survey

1.2.1 Research papers

The value of peer matching, mentorship, and flexible student support systems in overcoming social integration obstacles has been investigated by academics throughout the years. Though there are several options available, few combine interactive elements, adaptive relationship growth, and peer grouping into one platform. An overview of the major research studies that influenced this project may be found below:

1. **"The benefits of peer mentoring in higher education: findings from a systematic review"**[8] **-** Improvements in academic achievement, retention rates, mental health, social integration, and the significance of mentor-mentee compatibility in peer mentoring programs are all highlighted in this research. It discusses the difficulties new students have in unfamiliar academic settings and how peer mentorship eases the adjustment.
2. **"A Qualitative Study of Peer Relationships"** [9]**-** Examines the beneficial effects of peer connections on well-being, mental health, and organizational commitment that are important to peer matching systems.
3. **"Evaluating Similarity Variables for Peer Matching in Digital Health Communities"** [10]**-** The development of AI-driven compatibility algorithms can benefit from the discussion of peer matching algorithms that use similarity variables to promote constructive peer interactions.
4. **"Towards Personalized Peer-Matching for Learning"** [11]**-** Presents a useful implementation of academic peer review systems by presenting a personalized peer matching algorithm that distributes reviewers according to student requirements and performance.

5. **"The Nature and Evolution of the Mentoring Relationship in Academic Medicine"** [12] **-** Explains the stages of relationship evolution (cultivation, transition, and redefinition) in mentoring, which may help guide rematching reasoning and relationship progression.

6. **"Full article on matching groups that benefit students"** [13] **-** Discusses how group composition affects learning results and how it relates to group compatibility scores that are based on academic and emotional criteria.

7. **“Peer Review Market Design”** [14]**-** outlines a peer review market architecture that optimizes reviewer assignment and enhances review quality by integrating effort-based matching and admission control mechanisms. This is essential for successful peer matching systems in academic and professional contexts.

8. **“AI and peer reviews in higher education: students' multimodal views”** [15] **-** Investigates how students see AI-assisted peer reviews and feedback in higher education, pointing out both advantages and drawbacks while offering insightful information on how to include AI into peer matching systems.

9. **“Dynamic Matching Algorithms”** [16]**-** Explains the theoretical and practical features of dynamic matching algorithms for platforms that match users continually over time. These algorithms are important for creating rematching and adaptive peer matching systems.

10. **“Comparing human and AI expertise in the academic peer review process: towards a hybrid approach”** [17] - The advantages and disadvantages of both AI-based and human peer evaluations in academic publication are assessed in this study. It suggests a hybrid review methodology that enhances peer review quality by combining human contextual and ethical judgement with AI's efficiency.

1.3 Research Gap

Following table highlights gaps in existing studies and how proposed system addresses:

Table 1.1: Table that shows gap in existing studies and proposed solution

Features / Papers	Peer Mentoring Benefits [8]	Personalized Peer Matching [11]	Peer Review Market [14]	Mentoring Relationship Evolution [12]	AI Matching Algorithm [10]	UniPulse Features
Academic background matching	Partial (focus on mentoring)	Yes (performance-based)	No	Indirect (relationship stages)	Yes	Yes
Shared interests	No	No	No	No	Yes	Yes
Emotional traits and communication preference	No	No	No	Yes (relationship stages)	Partial	Yes
Group compatibility scoring	Yes - Indirect (well-being focus)	No	No	No	No	Yes
Dynamic relationship evolution & re-matching	No	No	Partial (effort-based)	Yes (cultivation to redefinition)	No	Yes
Feedback loop integration	Yes	No	No	No	No	Yes

With limited emphasis on ongoing assessment or dynamic rematching, most of the current peer matching and mentoring research concentrate on specific elements like academic alignment or relationship phases, depending on static and single-dimension compatibility. By implementing an intelligent peer matching approach that takes into consideration communication styles, emotional characteristics, shared interests, and academic background, our suggested system, UniPulse, fills this gap. Relationships are dynamically adjusted through compatibility scoring, feedback, and rematching logic. Through the integration of these elements into an effective framework, UniPulse offers a comprehensive, ever-changing support system that improves student achievement, well-being, and adaptability.

1.4 Research Problem

Who is affected by the problem?

Effective peer group formation is a difficulty for educators and learners using digital learning platforms since current methods frequently rely on self-selection or random allocation without considering the varied abilities, objectives, and learning styles of leaners.

What is the nature of the problem?

Peer groups that are unbalanced due to mismatched skill levels, interests, and availability lower cooperation quality, restrict engagement, and have negative impacts on overall learning outcomes. This is caused by the lack of an organized and intelligent peer matching system.

What could be done to address the problem?

To improve the effectiveness of collaborative learning and learner satisfaction, an Intelligent Peer Matching system that uses a weighted scoring method based on several factors (academic performance, subject knowledge, learning style, interests, and availability) could be developed and put into use. This system would guarantee balanced, relevant, and productive peer connections.

2. OBJECTIVES

2.1 Main Objective

To design and implement within one year an Intelligent Peer Matching system for university freshers using a weighted scoring method that ensures effective, balanced, and personalized peer group formation, while being continuously evaluated and reviewed for its impact on engagement, collaboration, and academic performance.

2.2 Specific Objectives

- **Compatibility-Based Matching Algorithms -** Implementation and use algorithms that examine many student characteristics, including academic background, common interests, emotional characteristics, and communication preferences. The objective is to pair university freshmen with peers who best fit freshmen's social requirements and learn preferences by developing specific and customized peer matches.

- **Group Compatibility Scoring System -** In this component, a systematic scoring model that assesses the general well-being of peer groups is developed. Several criteria, including academic alignment, active involvement, and emotional diversity, will be used to determine the score. Using this approach, the system is able to measure group dynamics and determine whether groups are balanced and conducive to students' academic and personal development.

- **Relationship Evolution and Re-Matching Mechanism -** The development of adaptive logic, which enables the system to dynamically restructure peer connections, is the focus of this sub-objective. The algorithm will suggest rematching with more appropriate peers when student compatibility declines. This

guarantees that partnerships continue to be fruitful and consistently in line with the changing demands of the students.

- **Feedback Loop, Visualization, and Gamification -** This part of the project integrates a continuous feedback mechanism where students share experiences after peer interactions. The system will use information to create network diagrams that show the strength and connectedness of social ties. To boost student motivation, promote involvement, and maintain engagement, gamified elements will also be implemented.

3. METHODOLOGY

3.1 Software Solution

To assist university students in finding appropriate peers based on students' academic background, hobbies, personality traits, and well-being profiles, UniPulse - AI-driven peer matching platform is the suggested software solution. To determine which fellow students are most likely to connect positively on a social, academic, and emotional level, the system analyses each student's survey replies and determines compatibility score with other students. Students can use the platform, which offers a straightforward and user-friendly interface for connecting with possible classmates.

A thorough survey that covers social and personality qualities, academic and demographic information, hobbies and lifestyle, wellness, and assistance requirements is initially filled out by students. Peer suggestions and compatibility calculations are based on these answers. A weighted scoring methodology is used to process each response, giving it a numerical value. The compatibility score is influenced by more significant aspects. This guarantees that while choosing peer matches, factors like personality, hobbies, academic alignment, and welfare are all properly considered.

Compatibility scores are determined by the platform by comparing each student's profile to that of other students and adding together the weighted responses from every survey category to generate a final percentage score. The weighted scoring weights are displayed in table 3.1 below. Additionally, students can filter possible classmates according to chosen social interaction style, academic subject, or common hobbies. This enables people to give priority to relationships that best match students' objectives, inclinations, and level of social comfort.

Weekly check-ins are included to provide information on peer interaction frequency, group participation, stress levels, and mood. The platform allows users to dynamically modify compatibility ratings because of these updates, guaranteeing that peer suggestions consider students' present emotional and social conditions. By displaying compatibility percentages and emphasizing common interests and personality traits, the site shows students the most suitable peer matches. To promote social integration, academic collaboration, and emotional health during college time, students can then decide to communicate with classmates through study groups, group activities, or one-on-one conversations. Student will able to communicate via the portal after peer recommendation.

Table 3.1: Weighted Scoring Model

Category	**Weight (%)**	**Sub-Factors**
Interest & Hobby Match	25%	Top Interests / Preferred Learning Style/ Participation in extracurriculars
Academic Compatibility	20%	Faculty / AL Stream
Personality Compatibility	20%	Introvert-Extrovert Scale / Comfort with Group Work / Stress/Anxiety Level / Weekly mood scale
Emotional & Wellbeing Alignment	15%	Overwhelmed/Anxious / Struggle to connect with peers / Mood similarity (weekly check-in emotion dropdown)
Communication Preference	10%	Preferred Communication Tools / Preferred Support Methods
Social Preferences	10%	Social Preference / Group vs. 1-on-1 participation / Peer interaction frequency (weekly check-in)

3.2 System Architecture

The system architecture is a student-centric structure intended to support the learner's journey holistically, as seen in figure 3.1. Fundamentally, the learner engages with four main management modules: Adaptive Learning and Student Profiling Individual management, conversational support and student wellbeing management, social inclusion risk detection and early intervention management, intelligent peer matching and relationship evolution management. A primary AI analysis engine that analyses data powers these modules, which don't function independently. This engine keeps track of and retrieves information from a central database, establishing an ongoing feedback loop. By ensuring that each contact and intervention is guided by thorough, data-driven insights, this architecture enables a highly customized and proactive support system that adjusts to the academic and social-emotional needs of the student.

Figure 3.1: High-level Diagram for UniPulse

As shown in Figure 3.2, AI-powered peer matching tool named UniPulse assists college students in finding compatible peers by examining students' profiles of well-being, hobbies, academic background, and personality features. A comprehensive survey spanning academic, social, lifestyle, and wellness topics is initially filled out by students. The results are then analyzed using a weighted scoring system to produce compatibility scores with other students. The platform prioritizes meaningful interactions by highlighting shared interests, academic alignment, and personality attributes. It also allows users to filter by social style, subject, or hobbies. Compatibility scores are dynamically updated to reflect students' current social and emotional states by using weekly check-ins that monitor interaction frequency, participation, stress, and mood. UniPulse promotes academic cooperation, social bonding, and emotional well-being throughout the university experience through study groups, events, or individual interactions.

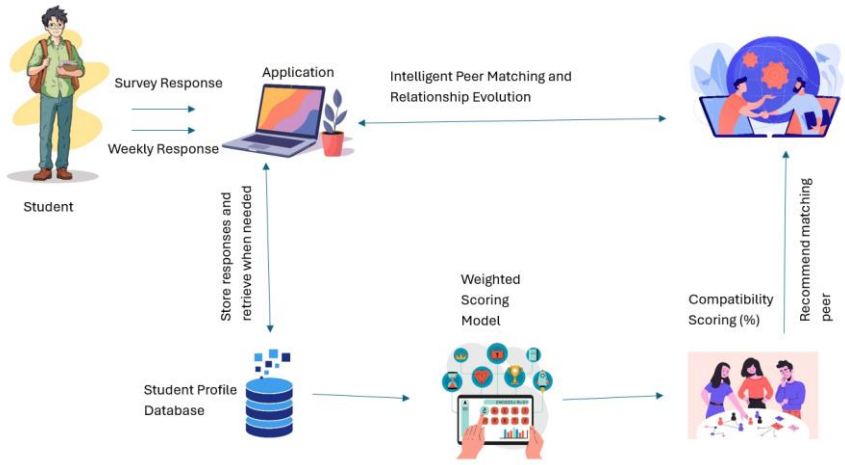

Figure 3.2: High-Level Diagram for Intelligent peer matching and relationship evolution platform

3.3 Data Availability

3.3.1 Ethical clearance

Ethical clearance for this research project, titled "AI-Based Social Integration and Inclusion Platform for University Freshers," has been secured through the SLIIT Ethics Review Committee's expedited procedure. The study involves collecting sensitive, individually identifiable data from first year, first semester students through comprehensive surveys, onboarding profiles, and weekly emotional check-ins. All data collection will proceed only after obtaining explicit informed consent, with transparent communication about usage protocols and privacy safeguards. Individually identifiable data will be encrypted and stored on secure institutional servers with rigorous access controls. Upon project completion, all raw data containing personal identifiers will be permanently deleted from all systems and devices to ensure participant privacy and confidentiality.

The intelligent peer matching and relationship evaluation management component of the platform will allow students to view recommended peer profiles generated through the system's compatibility analysis. Students can send a connection request after looking over a profile, and if it is accepted, they can use the platform to start a conversation. Additionally, by taking feedback and engagement levels into account, the system will continuously assess the quality of peer relationships, allow adaptive rematching when needed to guarantee meaningful and valuable student interactions. All raw chat logs and related identifiable records will be permanently removed from the system at the end of the project. This protects privacy and upholds ethical integrity by guaranteeing that no personal information will be available when the project is completed.

4. PROJECT REQUIREMENTS

To ensure the proposed peer matching platform's functionality, effectiveness, and dependability, project needs must be carefully determined. System specifications, functional and non-functional elements, and a methodical implementation strategy using a Gantt chart and work breakdown structure are all part of these criteria.

4.1 Data Collection

Since the matching algorithm depends on accurate and reliable student input, data collecting is an essential component of the project. Students' academic, social, personality, and wellness data will be gathered through surveys and weekly check-in forms. The weighted scoring methodology that determines peer compatibility scores will employ this data, which will be kept in a central database.

4.1.1 Functional requirements

The functional requirements define the essential operations and services that the UniPulse – Intelligent Peer Matching and Relationship Evolution Platform, must deliver to meet its objectives.

1. **Compatibility Scoring Engine -** Peer compatibility scores must be determined by the system using a weighted scoring model that considers communication preferences, personality traits, academic background, hobbies, and emotional health.

2. **Peer Recommendation Module -** According to compatibility percentage, the platform must recommend a list of the students' most compatible classmates.

3. **Filtering & Search Functionality -** Potential peers must be filtered by students according to preferences for communication methods, learning styles, hobbies, and instructors.

4. **Dynamic Re-Matching -** To recommend new peer matches as the quality of current partnerships decreases, the system must track peer interactions and update compatibility scores on a regular basis.

5. **Peer Group Formation -** To guarantee balanced intellectual and emotional alignment, the platform must enable the formation of peer groups (study groups, support circles) by bringing together many students who are very compatible.

4.1.2 Non-functional requirements

The non-functional requirements define the quality attributes and performance standards that ensure UniPulse is reliable, secure, and scalable.

- **Performance -** To guarantee real-time peer recommendations, the system completes compatibility calculations for a minimum of 10,000 students in a matter of seconds.
- **Scalability -** The system is scalable from faculty-level experiments to large universities with 50,000+ users, allowing for institution-wide implementation.
- **Usability -** Students with different levels of digital literacy should be able to easily navigate the user interface and utilize it on mobile devices.
- **Reliability & Availability -** To prevent data loss, the platform should provide disaster recovery and automated backup features in addition to 99.9% uptime.
- **Maintainability -** Modularity in the system would enable seamless modifications to the survey questions, matching algorithms, and gamification components.

4.2 System Requirements

4.2.1 Software requirements

Laravel 11 will serve as the main backend framework for the system's development because of its inherent security features, scalability, and resilience. The Model-View-Controller (MVC) architecture of Laravel will guarantee a clear division of responsibilities and enhance maintainability. HTML, CSS, and JavaScript will be used in the frontend design to create a responsive and intuitive user experience. To support features like mailing, authentication, and API processing, more Laravel packages and libraries will be incorporated. In the beginning, a local development environment like XAMPP or Laravel Sail will be utilized, and a cloud-based hosting provider will be utilized to manage deployment.

4.2.2 Hardware requirements

To ensure availability and performance, the platform needs a reliable and secure hosting environment with cloud-based servers. To handle heavy processing loads and quick data retrieval, servers with multi-core processors, SSD storage, and at least 8 GB of RAM are recommended specifications. To engage with the system via a web browser or mobile application, end users just need simple technology, such as smartphones, tablets, or PCs with internet connectivity.

4.2.3 Database management

User information, preferences, and compatibility scores will be stored and managed using a relational database management system (RDBMS), such as MySQL. To guarantee effectiveness and dependability, appropriate data normalization, indexing, and query optimization strategies shall be used. To preserve data confidentiality and integrity and safeguard sensitive information, access controls and encryption techniques will be implemented. To prevent data loss, regular backups and recovery procedures will also be included.

4.2.4 AI Model integration

Through the analysis of user data and the discovery of significant trends, the AI model integration aims to improve the platform's capacity to provide intelligent and customized services. More accurate peer matching and compatibility ratings are made possible by the system's ability to group users with comparable academic interests, hobbies, or personality qualities using clustering and recommendation approaches. To ensure real-time processing of survey replies and profile updates, the AI model will be embedded into the backend as a service, enabling smooth contact with the database and user interface. As the model continuously learns from fresh data to enhance recommendations and user happiness over time, this integration not only increases match accuracy but also gives the system flexibility.

4.2.5 Version controlling

To guarantee cooperative and effective work, Git and platforms like GitHub will be used to maintain version control. Developers will be able to easily integrate updates, manage several branches, and keep track of changes through this tool. By maintaining an orderly development history and enabling rollback capabilities in the event of errors, proper version control procedures will enhance project management and reduce deployment risks.

4.3 Work Break Down Structure

For efficient planning and execution, the project's primary tasks and subtasks are broken down into manageable components in the Work Breakdown Structure (WBS) that follows.

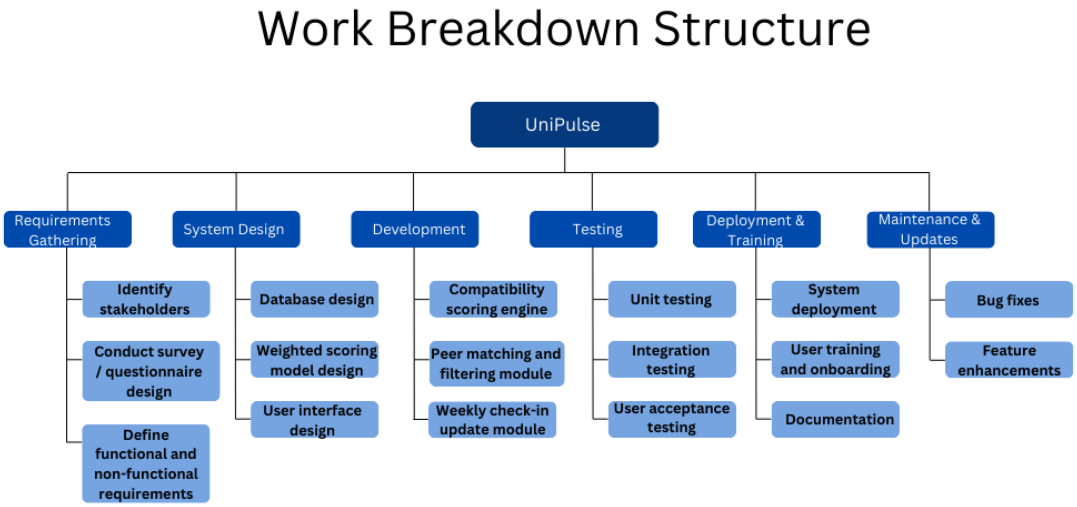

Figure 4.1: Work Breakdown Structure for UniPulse

4.4 Gantt Chart

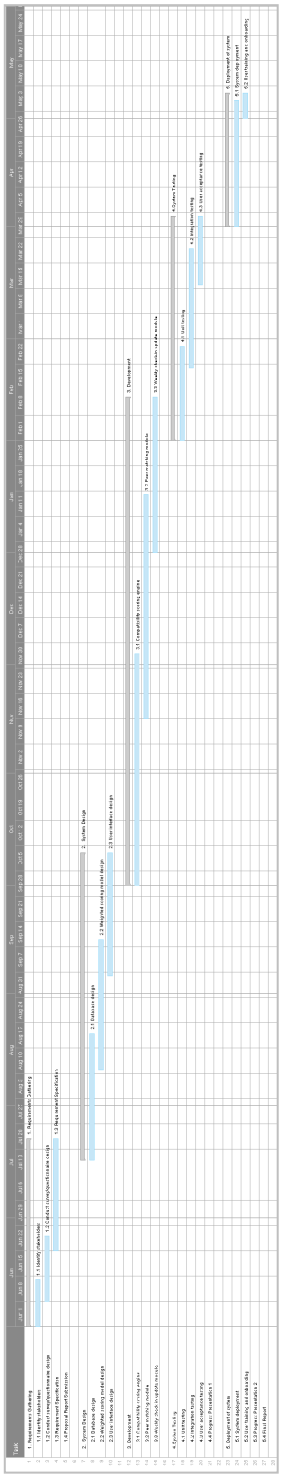

Figure 4.2: Gantt Chart for UniPulse

For efficient project management, the Gantt chart below shows the project timetable, emphasizing important activities, durations, dependencies, and milestones.

The Smartsheet Gantt chart is available at the following website for thorough task tracking and real-time updates:
https://app.smartsheet.com/sheets/x9hmhfRQwvccHC6vVxgGQqv9V79Q8xMQvXrW7hF1

5. BUDGET JUSTIFICATION

The UniPulse AI-Based Social Integration and Inclusion Platform is expected to cost Rs. 725,000.00 in total. To guarantee strong development, seamless acceptance, and excellent usability, the budget is distributed among certain operational, technical, and human resource components. System development, interface and design, personnel expenses, testing, and support activities are all included in the breakdown.

Table 5.1: Budget Justification for UniPulse

Cost Area	Description	Estimated Cost (Rs.)
Backend Development	Building the platform's main system, AI features, and database.	180,000.00
Frontend Development	Creation of responsive and interactive user interfaces enhancing usability and accessibility for university freshers.	80,000.00
UI/UX Design & Prototyping	Design layouts, wireframes, prototyping, user testing.	90,000.00
Developers & Technical Support Staff	Salaries for developers, AI engineers, support staff.	150,000.00
Project Management	Planning, coordination, and documentation.	80,000.00
Training & Awareness Workshops	Conducting awareness workshops and providing user guides for students and staff.	30,000.00

AI Model and Algorithm	Developing AI models and algorithms	25,000.00
Testing & Quality Assurance	Functional testing, bug fixing, performance validation	40,000.00
Maintenance & Minor Updates	Post-deployment fixes, small improvements	50,000.00
Total Estimated Cost		**725,000.00**

6. COMMERCIALIZATION ASPECT

The commercialization of UniPulse is centered on positioning it as a scalable, cost-effective wellbeing and integration platform for universities. The primary market consists of universities facing high dropout rates and increasing demand for mental health support, particularly for first year and international students, introverts, and students from rural or unfamiliar environments who may fear organizations or struggle to adjust. These groups are most vulnerable to isolation, stress, and disengagement, making them the core focus of UniPulse. Extroverted students may also benefit if students choose to use the system for self-reflection and support, though they are not the primary target. UniPulse will be marketed through direct institutional partnerships, pilot projects, and presence at educational technology forums, highlighting its ability to reduce attrition and enhance student satisfaction.

Implementation follows a phased approach, beginning with a faculty-level pilot to demonstrate impact and gather feedback, followed by institution-wide rollout and eventual integration with the university's LMS and student portals. This staged deployment ensures smooth adoption while allowing the platform to adapt to the institution's unique needs. The financial model is designed around a subscription-based SaaS framework, where universities pay either a yearly license fee or a per-student subscription. Tiered pricing options will allow institutions to choose between core wellbeing monitoring and premium analytics with counselor integration.

From a financial perspective, UniPulse offers universities clear returns by reducing student attrition, which directly preserves tuition revenue while also enhancing institutional reputation. For example, even a small reduction in dropout rates translates into significant financial savings and improved rankings. Operating costs mainly cover AI system maintenance, cloud hosting, data security, and customer support, making the model sustainable at scale. UniPulse's competitive edge lies in its proactive and AI-

driven nature: unlike traditional counseling services that react after problems occur, the platform continuously monitors trends, provides personalized support, and escalates serious cases automatically. This ensures universities can extend mental health and wellbeing support to every student, at scale, without proportional increases in staffing costs.

7. CONCLUSION

The Peer Matching System is a complete platform that aligns personality traits, communication preferences, common interests, and academic compatibility to create meaningful connections between students. The method guarantees that peer suggestions are based on defined, data-driven compatibility criteria rather than being made at random by utilizing a weighted scoring model. Students may work together more successfully, exchange knowledge, and form helpful networks in this setting.

The method prioritizes total student participation by taking social and emotional compatibility into consideration in addition to academic collaboration. This harmony between personal and academic aspects improves learning results and the general wellbeing of students. The platform ensures user confidence by guaranteeing accessibility and dependability through specified criteria, secure data collecting, and an intuitive design.

In conclusion, the Peer Matching System provides a scalable, flexible, and safe answer to the contemporary problems associated with student integration. It gives students the chance to create enduring relationships outside of the classroom, which eventually promotes community development and personal improvement. Institutions stand to build more robust, cooperative, and emotionally nurturing learning environments when students use such systems.

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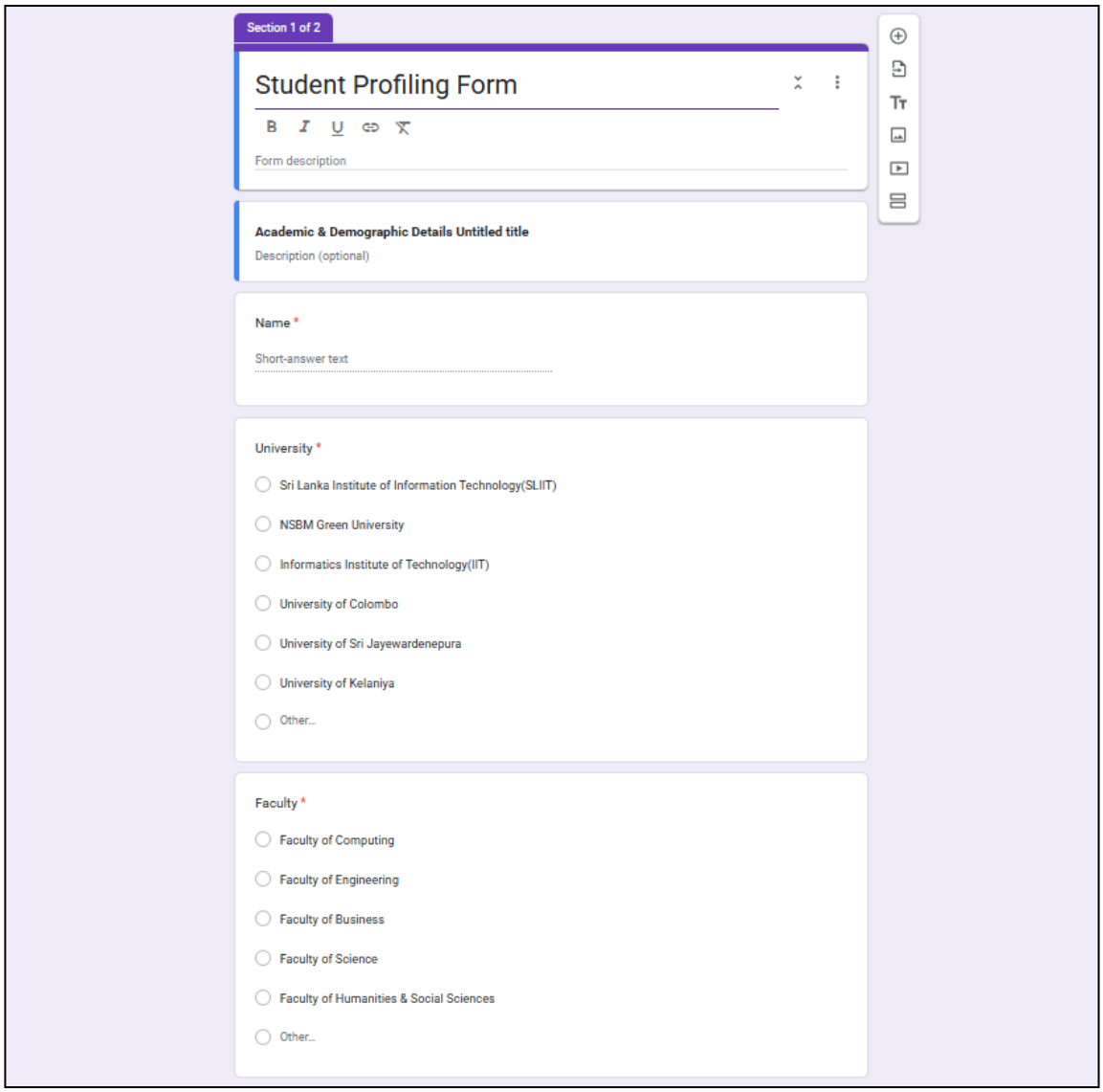
Appendices

Appendix A: Input Form 1



AL Stream *

- Bio Science
- Physical Science
- Arts
- Commerce
- Other...

AL Results *

	A	B	C	S	F
Subject 1					
Subject 2					
Subject 3					
English					
General Knowl...					

What is your preferred learning style *

- [ ] Online
- [ ] Physical
- [ ] Hybrid

How confident are you in transitioning to university life? *

	1	2	3	4	5	
Not confident at all						Extremely confident

Social & Personality Traits

Description (optional)

Which social setting do you prefer most? *

- 1-on-1 interactions
- Small Groups
- Large Groups
- Online-only
- Other...

Where would you place yourself on the Introvert–Extrovert scale? *

	1	2	3	4	5	6	7	8	9	10	
Fully Introvert	○	○	○	○	○	○	○	○	○	○	Fully Extrovert

In general, how would you describe your usual level of stress or anxiety? *

- Low
- Moderate
- High

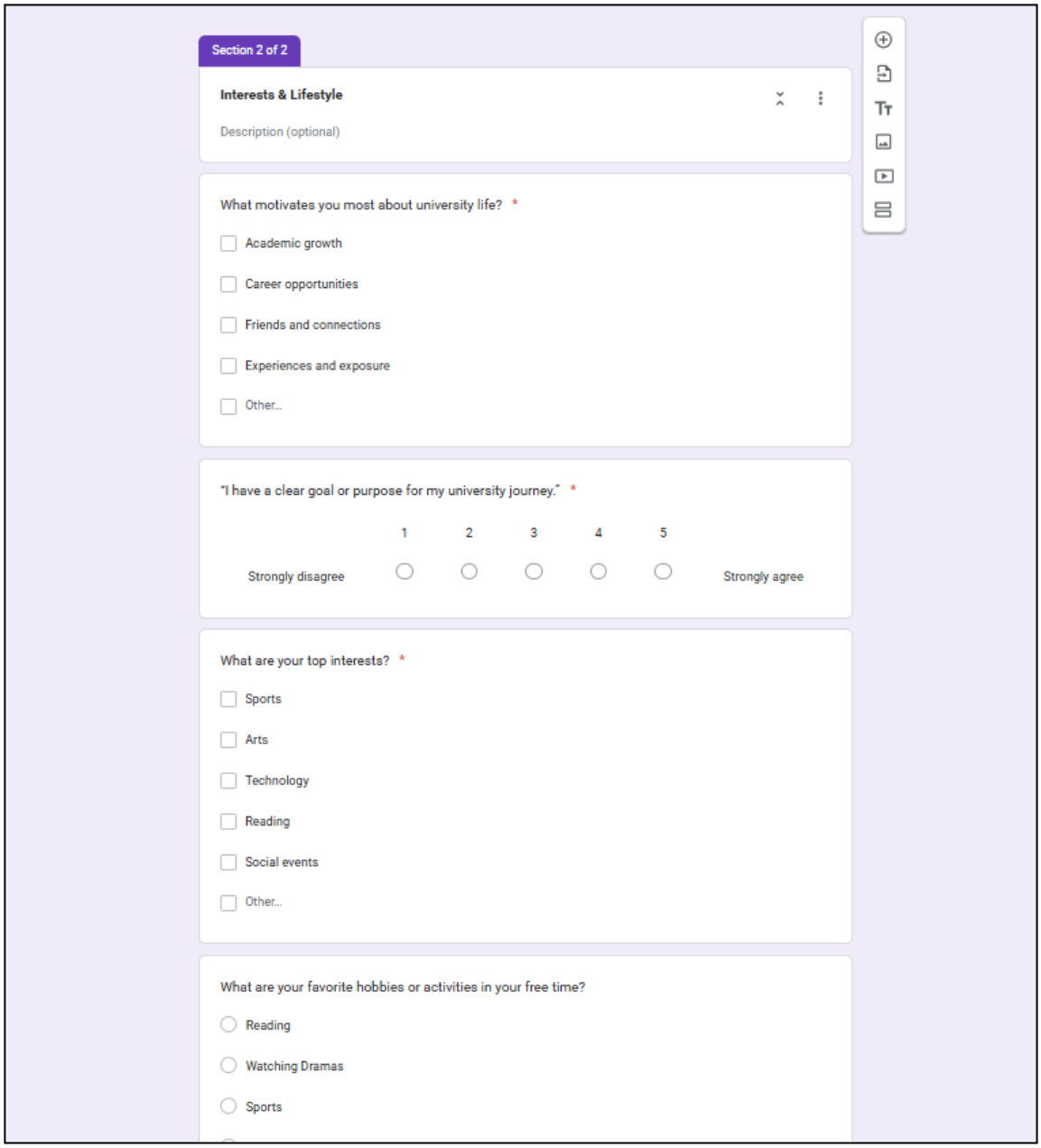
How comfortable are you working in group activities? *

	1	2	3	4	5	
Not comfortable at all	○	○	○	○	○	Very comfortable

Which communication methods do you prefer? (Select all that apply) *

- [ ] Texts
- [ ] Calls
- [ ] In-person conversations

After section 1 Continue to next section



Section 2 of 2

Interests & Lifestyle

Description (optional)

What motivates you most about university life? *

- [ ] Academic growth
- [ ] Career opportunities
- [ ] Friends and connections
- [ ] Experiences and exposure
- [ ] Other...

"I have a clear goal or purpose for my university journey." *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

What are your top interests? *

- [ ] Sports
- [ ] Arts
- [ ] Technology
- [ ] Reading
- [ ] Social events
- [ ] Other...

What are your favorite hobbies or activities in your free time?

- ○ Reading
- ○ Watching Dramas
- ○ Sports

What are your favorite hobbies or activities in your free time?

- Reading
- Watching Dramas
- Sports
- Painting
- Traveling
- Volunteering
- Gaming
- Listening to music
- Other...

What is your current living arrangement? *

- Hostel
- Home
- Boarding place
- Other...

Are you currently employed? *

- Yes
- No

Wellbeing & Support Needs Untitled title

Description (optional)

"I often feel overwhelmed or anxious." *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

Other...

Are you currently employed? *

Yes

No

Wellbeing & Support Needs Untitled title

Description (optional)

"I often feel overwhelmed or anxious." *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

"I struggle to connect with peers." *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

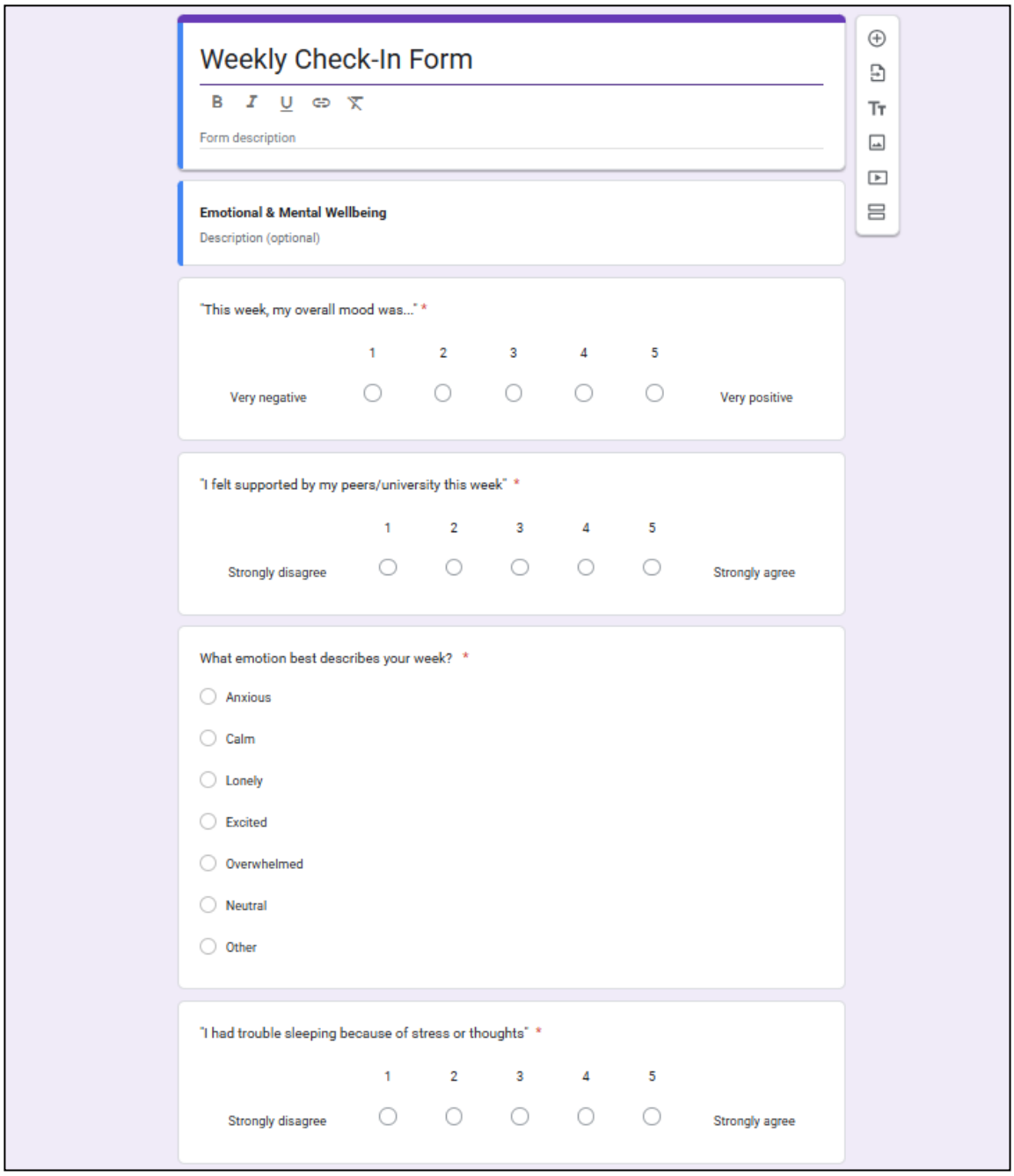
"I would use an AI platform for wellbeing support." *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

Which types of support would you find helpful? *

- [ ] Peer matching
- [ ] Counseling services
- [ ] Study groups
- [ ] AI chatbot support

Appendix B: Input Form 2



Weekly Check-In Form

Form description

Emotional & Mental Wellbeing

Description (optional)

"This week, my overall mood was..." *

	1	2	3	4	5	
Very negative	○	○	○	○	○	Very positive

"I felt supported by my peers/university this week" *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

What emotion best describes your week? *

- ○ Anxious
- ○ Calm
- ○ Lonely
- ○ Excited
- ○ Overwhelmed
- ○ Neutral
- ○ Other

"I had trouble sleeping because of stress or thoughts" *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

Social & Academic Behavior Untitled title

Description (optional)

How often did you interact with peers outside class this week? *

	1	2	3	4	5	
Not at all	○	○	○	○	○	Daily

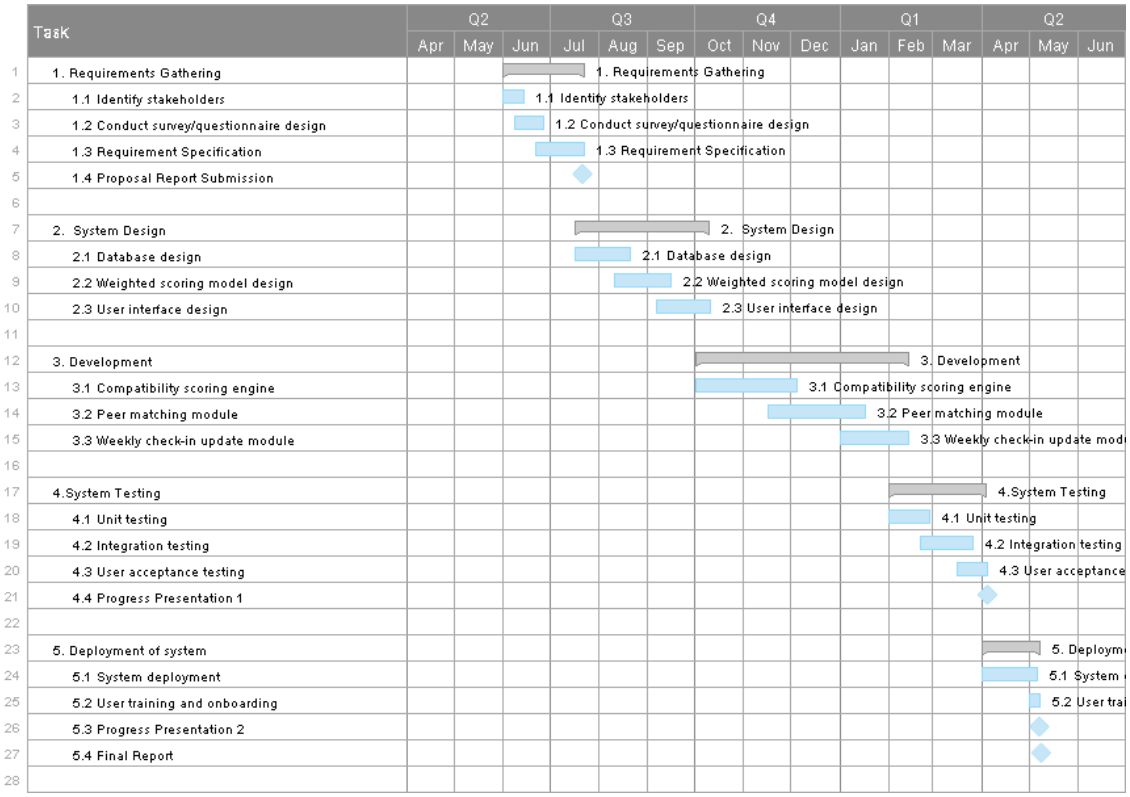
"I was able to keep up with my academic workload this week" *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

Which group activities did you participate in this week? *

- [ ] Study groups
- [ ] Social events
- [ ] Clubs
- [ ] None

What academic challenges did you face this week? *

- ○ Assignments
- ○ Exams
- ○ Group work
- ○ Understanding lessons
- ○ Time management
- ○ Other
- ○ None

Appendix C: Project Timeline with Milestones